Highlights

Resonant Frequencies of the Human Urinary Bladder:
A Fluid-Filled Viscoelastic Shell Model

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- Bladder n=2 flexural mode at 12–18 Hz across physiological fill states
- Frequency minimum (∼12 Hz) at 150–200 mL due to competing geometry and stiffening
- Mechanical (WBV) coupling exceeds airborne acoustic coupling by ∼7,600×
- Same analytical framework as abdominal cavity model, with fill-dependent parameters
- Results consistent with occupational reports of vibration-induced urinary urgency

Resonant Frequencies of the Human Urinary Bladder: A Fluid-Filled Viscoelastic Shell Model

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Abstract

Occupational exposure to whole-body vibration (WBV) is associated with increased urinary urgency among vehicle operators, yet no first-principles model of bladder resonance exists to explain this observation. We model the human urinary bladder as a fluid-filled viscoelastic spherical shell whose geometry, wall stiffness, and internal pressure vary with fill volume (50 mL to 500 mL). The framework—an adaptation of an oblate spheroidal shell model developed for the abdominal cavity—predicts that the fundamental flexural mode ($n = 2$) lies between 12 Hz and 18 Hz, with a minimum near 12 Hz at approximately 170 mL. This minimum arises from the competition between increasing radius (which lowers frequency) and strain-stiffening of the detrusor muscle (which raises it). Coupling analysis shows that mechanical vibration transmitted through the pelvis is approximately 7,600 times more effective than airborne sound at exciting bladder wall modes, consistent with the dominance of the WBV pathway reported in the epidemiological literature. The predicted frequency range sits above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz) but within the broader WBV-affected spectrum, suggesting a forced sub-resonant response mechanism rather than direct resonance.

Keywords: bladder resonance, fluid-filled shell, whole-body vibration, viscoelastic shell, urinary urgency, ISO 2631

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[1] Large language model. Performed literature review, analytical computation, code development, figure generation, and manuscript drafting under human supervision.

1. Introduction

Whole-body vibration (WBV) at 4 Hz to 16 Hz is a recognised occupational hazard for vehicle operators. ISO 2631-1 [1] documents peak pelvic transmissibility at 4 Hz to 8 Hz, and epidemiological studies report a consistent association between prolonged WBV exposure and lower urinary tract symptoms—particularly increased urgency and frequency—among truck drivers, bus operators, and forklift operators [2, 3]. Despite this well-established empirical record, no first-principles mechanical model of the urinary bladder exists to explain *why* whole-body vibration should preferentially excite this organ, or to predict the frequency range at which the bladder wall is most susceptible to vibratory loading.

The urinary bladder is, from a structural dynamics perspective, a remarkably clean example of a fluid-filled viscoelastic shell. The detrusor muscle forms a roughly spherical wall of thickness 1.5 mm to 5 mm (depending on fill state) enclosing urine—an incompressible fluid with density and bulk modulus essentially identical to water. The geometry, wall stiffness, and internal pressure all vary systematically with fill volume, making the bladder a natural candidate for parametric analysis across its physiological operating range.

In a companion study [4], we developed an oblate spheroidal shell model of the human abdominal cavity to investigate the coupling between airborne infrasound and gastrointestinal resonance. That analysis revealed a coupling disparity of approximately 6.6×10^4 between mechanical and airborne excitation pathways—a result that explains why whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects while airborne infrasound at comparable frequencies does not. The present work applies the same analytical framework to the bladder, adapting the geometry, material properties, and boundary conditions to this smaller, thinner-walled, and more compliant organ.

The mechanical properties of the bladder wall have been characterised by several groups. Ultrasound bladder vibrometry [5] measured shear moduli of 9.6 kPa to 48.7 kPa across fill states from 187 mL to 267 mL, demonstrating

pronounced strain-stiffening. Barnes [6] performed tensile testing of excised bladder tissue, reporting Young's moduli from 10 kPa at rest to several hundred kilopascals at high stretch ratios. The wall exhibits significant viscoelastic behaviour: stress relaxation, creep, and a loss tangent estimated at 0.3–0.5 from dynamic measurements [7]. Wall thickness decreases from approximately 5 mm when empty to 1.5 mm to 3 mm at capacity [8], consistent with a constant tissue volume assumption. Intravesical pressure follows the characteristic cystometric curve: a compliant phase at low fill (5 cm to 10 cm water column) giving way to steep pressure rise above 300 mL [9].

We apply these data to the fluid-filled shell framework. The bladder is modelled as a spherical shell (the near-spherical approximation is appropriate here, unlike the distinctly oblate abdominal cavity) with fill-dependent radius, wall thickness, elastic modulus, and internal pressure. We compute the flexural mode frequencies across the physiological fill range (50 mL to 500 mL), identify the minimum-frequency fill state, and compare airborne acoustic coupling with mechanical WBV coupling to the bladder wall.

The remainder of this paper is organised as follows. Section 2 presents the mathematical formulation, including the fill-dependent parameter model and the modal analysis. Section 3 reports flexural mode frequencies across fill volume, identifies the frequency minimum, and quantifies the coupling disparity between mechanical and airborne excitation. Section 4 discusses clinical implications, limitations, and comparison with the abdominal cavity model. Section 5 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human urinary bladder is idealised as a thin spherical shell of radius R, wall thickness h, completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Unlike the abdominal cavity—which requires an oblate spheroidal geometry with aspect ratio $c/a \approx 0.67$ [4]—the bladder is

approximately spherical at moderate-to-full fill states, making the spherical shell approximation both simpler and more appropriate.

The thin-shell assumption requires $h/R \ll 1$. For the bladder at 300 mL, $h \approx 1.8$ mm and $R \approx 42$ mm, giving $h/R \approx 0.04$, which satisfies the condition comfortably. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is therefore applicable.

The bladder wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{1}$$

where η is the loss tangent. The quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2$. Published values of η for bladder wall tissue range from 0.3 to 0.5 [7, 6]; we adopt $\eta = 0.4$ ($Q = 2.5$, $\zeta = 0.2$) as a central estimate. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1 - \nu^2)}. \tag{2}$$

2.2. Fill-Dependent Parameters

All geometric and material properties vary with fill volume V (mL). We adopt the following constitutive relationships, drawn from the literature reviewed in §1.

Radius.. The equivalent sphere radius follows directly from the fill volume:

$$R = \left(\frac{3V}{4\pi}\right)^{1/3}, \tag{3}$$

giving $R = 2.3$ cm at 50 mL and $R = 4.9$ cm at 500 mL.

Wall thickness.. We assume approximately constant tissue volume V_{tissue}, calibrated to $h_0 = 5$ mm at $V_0 = 50$ mL. The tissue shell volume is

$$V_{\text{tissue}} = \frac{4}{3}\pi \left[(R_0 + h_0)^3 - R_0^3\right], \tag{4}$$

from which $h(V)$ is obtained by solving

$$(R + h)^3 = R^3 + \frac{3V_{\text{tissue}}}{4\pi}, \tag{5}$$

subject to a physiological floor of $h_{\min} = 1.5\,\mathrm{mm}$. This yields $h = 5.0\,\mathrm{mm}$ at 50 mL, thinning to 1.5 mm at 400 mL and above.

Elastic modulus.. Bladder wall stiffness increases dramatically with stretch (strain-stiffening), a hallmark of collagenous soft tissue [10]. We model this as a log-linear interpolation between vibrometry-derived bounds [5, 6]:

$$E(V) = E_{\min}\left(\frac{E_{\max}}{E_{\min}}\right)^{\phi}, \qquad \phi = \frac{V - V_{\min}}{V_{\max} - V_{\min}}, \tag{6}$$

where $E_{\min} = 10\,\mathrm{kPa}$ at $V_{\min} = 50\,\mathrm{mL}$ and $E_{\max} = 200\,\mathrm{kPa}$ at $V_{\max} = 500\,\mathrm{mL}$, with ϕ clipped to $[0, 1]$.

Intravesical pressure.. The pressure–volume relationship follows the characteristic cystometric curve [9]: a compliant phase at low fill followed by a steep rise. We model this as a quadratic:

$$P(V) = P_{\min} + (P_{\max} - P_{\min})\,\phi^2, \tag{7}$$

where $P_{\min} = 490\,\mathrm{Pa}$ ($\approx 5\,\mathrm{cmH_2O}$) and $P_{\max} = 2940\,\mathrm{Pa}$ ($\approx 30\,\mathrm{cmH_2O}$).

Table 1: Bladder model parameters at representative fill states.

Parameter	Symbol	50 mL	150 mL	300 mL	500 mL	Unit
Radius	R	2.29	3.30	4.15	4.92	cm
Wall thickness	h	5.00	2.73	1.79	1.50	mm
Elastic modulus	E	10.0	19.5	52.8	200.0	kPa
Intravesical pressure	P	5.0	6.2	12.7	30.0	$\mathrm{cmH_2O}$
Poisson's ratio	ν	0.49 (all fill states)				—
Wall density	ρ_w	1050				$\mathrm{kg/m^3}$
Fluid density	ρ_f	1020				$\mathrm{kg/m^3}$
Fluid bulk modulus	K_f	2.2				GPa
Loss tangent	η	0.40				—

Table 1 summarises the model parameters at four representative fill states. The Poisson's ratio $\nu = 0.49$ reflects the near-incompressibility of soft tissue.

The fluid properties correspond to urine at body temperature, which is effectively identical to water for acoustic purposes.

2.3. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation. As in the abdominal model [4], the restoring force is dominated by the fluid bulk modulus:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}}, \tag{8}$$

where $k_{\text{shell}} = 2Eh/[R^2(1-\nu)]$ and $k_{\text{fluid}} = 3K_f/R$. For the bladder at 300 mL, $k_{\text{fluid}} \approx 1.6 \times 10^{11}$ Pa/m while $k_{\text{shell}} \approx 2.2 \times 10^4$ Pa/m—a ratio exceeding 10^6. The breathing mode frequency is approximately 9500 Hz, firmly in the ultrasonic range and irrelevant to WBV exposure.

2.4. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes that preserve the enclosed volume to leading order. The fluid contributes only added mass, not volumetric stiffness. Following Lamb [11] and Junger & Feit [12], the natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}, \tag{9}$$

where the three stiffness contributions are:

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{10}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{11}$$

Pre-stress stiffness.. The intravesical pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{12}$$

The effective mass per unit area includes both the shell wall and the fluid added mass:

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{13}$$

For a bladder at 300 mL, the model predicts $f_2 = 13.4\,\text{Hz}$, $f_3 = 23.5\,\text{Hz}$, and $f_4 = 34.8\,\text{Hz}$. The $n = 1$ mode corresponds to rigid-body translation and has zero natural frequency for a free shell; it is therefore excluded from the analysis.

2.5. Airborne Acoustic Coupling

At $f_2 \approx 13\,\text{Hz}$ the acoustic wavelength in air is $\lambda \approx 26\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R}{c_{\text{air}}} \approx 0.010 \ll 1. \tag{14}$$

The bladder lies deep in the Rayleigh regime. In a multipole expansion of the incident plane wave, the n-th order component scales as [12]

$$p_{\text{eff}}^{(n)} = p_{\text{inc}} \times (ka)^n. \tag{15}$$

For the $n = 2$ mode:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 1.0 \times 10^{-4}. \tag{16}$$

Only 0.01% of the incident pressure drives the fundamental flexural mode via the airborne pathway.

2.6. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration transmits through the skeleton and pelvic floor as base excitation of the bladder wall. The coupling chain comprises two stages:

Seat-to-pelvis transmissibility.. ISO 2631 data indicate a pelvic resonance near $f_{\text{pelvis}} \approx 5.5\,\text{Hz}$ with $\zeta_{\text{pelvis}} \approx 0.25$. The transmissibility at frequency f is

$$T(f) = \sqrt{\frac{1 + (2\zeta_p r_p)^2}{(1 - r_p^2)^2 + (2\zeta_p r_p)^2}}, \qquad r_p = \frac{f}{f_{\text{pelvis}}}. \tag{17}$$

At $f_2 = 13.4\,\text{Hz}$, $T \approx 0.31$: the pelvic system attenuates the input, as we are above its resonance.

Bladder modal amplification.. The bladder wall response to base excitation is characterised by the single-degree-of-freedom frequency response:

$$H(f) = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{f}{f_n}. \tag{18}$$

At resonance ($r = 1$), $H = Q = 2.5$.

Coupling ratio.. The mechanical-to-airborne coupling ratio at the $n = 2$ resonance frequency is

$$\mathcal{R} = \frac{T(f_2) \times H(f_2)}{(ka)^2} \approx \frac{0.31 \times 2.5}{1.0 \times 10^{-4}} \approx 7{,}600. \tag{19}$$

Mechanical vibration transmitted through the pelvic skeleton is approximately four orders of magnitude more effective at exciting bladder wall modes than airborne sound.

3. Results

3.1. Flexural Mode Frequencies Across Fill Volume

Figure 1 presents the $n = 2$ and $n = 3$ flexural mode frequencies across the physiological fill range (50 mL to 500 mL). The $n = 2$ frequency follows a distinctive U-shaped curve, with a minimum of approximately 12 Hz near 170 mL and higher values at both extremes of the fill range.

The physical mechanism behind this non-monotonic behaviour is a competition between two opposing effects:

1. **Geometric softening.** As the bladder fills, the radius increases (equation 3), which lowers the natural frequency through the R^{-4} and R^{-2} dependence in the bending and membrane stiffness terms (equations 10–11). The wall also thins (equation 5), further reducing both bending rigidity ($D \propto h^3$) and membrane stiffness ($\propto h$).
2. **Strain-stiffening.** The elastic modulus of the detrusor muscle increases exponentially with fill volume (equation 6), from 10 kPa at 50 mL to 200 kPa at 500 mL. This twenty-fold increase eventually dominates the geometric effects.

At low fill volumes the geometric effect dominates and the frequency falls; at high fill volumes the strain-stiffening wins and the frequency rises. The crossover produces the minimum near 170 mL—a fill state corresponding to the early "need to go" sensation in most individuals.

Table 2 lists the computed frequencies at key fill states. The $n = 2$ mode spans 12 Hz to 18 Hz and the $n = 3$ mode spans 20 Hz to 31 Hz across the physiological range. All predicted frequencies lie above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz), though below 20 Hz they remain within the broader frequency range affected by whole-body vibration.

Table 2: Flexural mode frequencies at key fill volumes.

Volume (mL)	R (cm)	h (mm)	E (kPa)	f_2 (Hz)	f_3 (Hz)	f_4 (Hz)
50	2.29	5.00	10.0	16.2	27.0	39.8
100	2.88	3.46	13.9	12.9	21.5	31.1
150	3.30	2.73	19.5	12.1	20.5	29.8
200	3.63	2.30	27.1	12.1	20.9	30.6
300	4.15	1.79	52.8	13.4	23.5	34.8
400	4.57	1.50	102.8	15.3	27.1	40.3
500	4.92	1.50	200.0	17.8	31.3	46.3

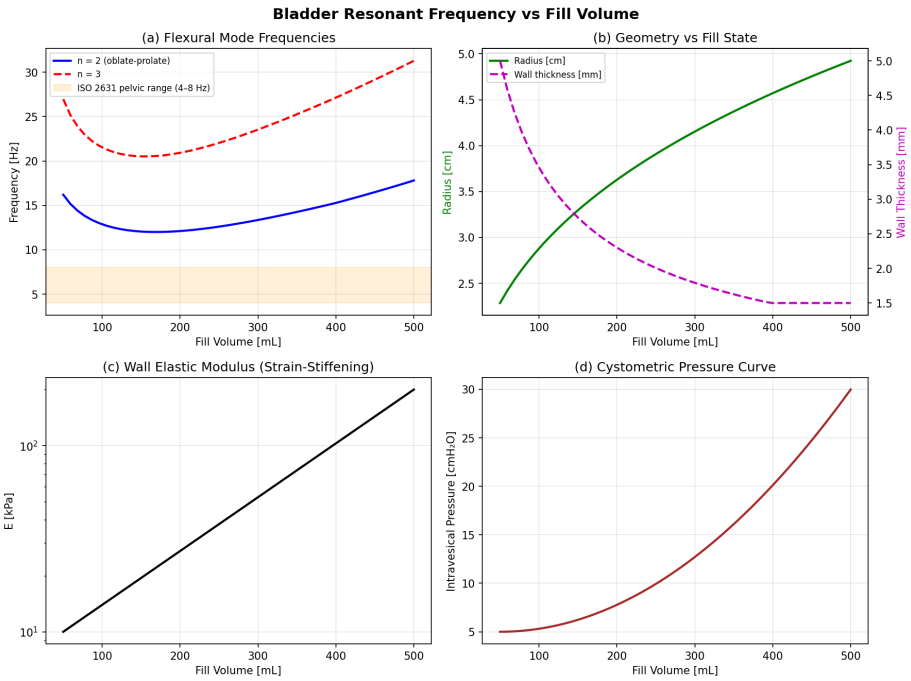

Figure 1: Bladder resonant properties across fill volume. (a) Flexural mode frequencies f_2 and f_3, with the ISO 2631 pelvic resonance band (4 Hz to 8 Hz) shown for reference. (b) Geometry: equivalent sphere radius and wall thickness. (c) Wall elastic modulus showing exponential strain-stiffening. (d) Intravesical pressure following the cystometric curve. The minimum $f_2 \approx 12$ Hz occurs near 170 mL, where geometric softening and strain-stiffening are in balance.

3.2. Coupling Pathways

Figure 2 compares the airborne and mechanical coupling pathways to the bladder at a representative fill of 300 mL ($f_2 = 13.4$ Hz). Three key observations emerge.

First, the airborne acoustic coupling scales as $(ka)^2$ and is vanishingly small across the entire frequency range of interest: at 13.4 Hz, $(ka)^2 \approx 1.0 \times 10^{-4}$. The bladder, like the abdominal cavity, is deep in the Rayleigh regime at all physiologically relevant frequencies.

Second, the mechanical coupling—the product of seat-to-pelvis transmissibility and bladder modal amplification—peaks near the bladder resonance frequency. Even though pelvic transmissibility is attenuated at 13.4 Hz ($T \approx 0.31$), the bladder modal amplification ($Q = 2.5$) partially compensates.

Third, the coupling ratio $\mathcal{R} \approx 7{,}600$ at f_2 (equation 19) confirms that mechanical vibration is the dominant excitation pathway by nearly four orders of magnitude. This is quantitatively consistent with—though somewhat smaller than—the 6.6×10^4 ratio found for the abdominal cavity [4], the difference arising primarily from the higher f_2 (and hence smaller Q contribution from pelvic transmissibility) of the bladder.

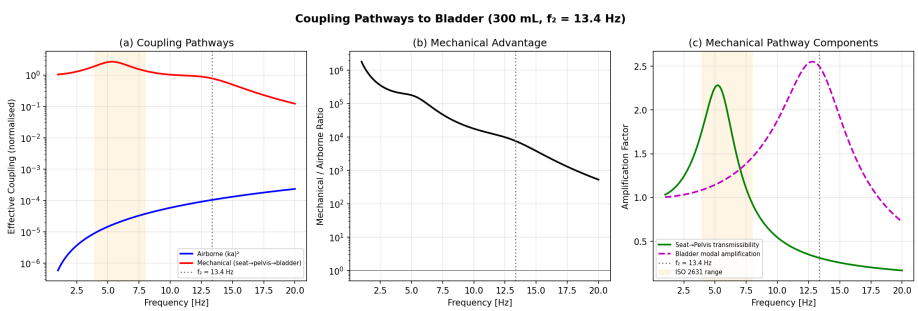

Figure 2: Coupling pathways to the bladder at 300 mL fill. (a) Airborne $(ka)^2$ coupling (blue) vs mechanical seat–pelvis–bladder coupling (red), with the n=2 resonance frequency marked. (b) Ratio of mechanical to airborne coupling (mechanical advantage), reaching $\sim 10^4$–10^6 across the frequency range. (c) Components of the mechanical pathway: seat-to-pelvis transmissibility (green) and bladder modal amplification (magenta).

3.3. Comparison with the Abdominal Cavity

Table 3 compares key quantities between the bladder and abdominal cavity models. The bladder is smaller, softer, and more heavily damped, yet the two organs share the same qualitative physics: fluid-dominated breathing modes at kilohertz frequencies, low-frequency flexural modes governed by the competition between shell stiffness and fluid added mass, and a severe airborne coupling penalty in the Rayleigh regime.

Table 3: Comparison of bladder and abdominal cavity models.

Quantity	Bladder (300 mL)	Abdomen
Equivalent radius	4.2 cm	15.7 cm
Wall thickness	1.8 mm	10 mm
Elastic modulus	53 kPa	100 kPa
Poisson's ratio	0.49	0.45
Loss tangent	0.40	0.25
Quality factor Q	2.5	4.0
Breathing mode f_0	∼9500 Hz	∼2500 Hz
Flexural f_2	13.4 Hz	4.0 Hz
ka at f_2	0.010	0.011
Coupling ratio $\mathcal{R}$	∼7,600	∼66,000

The principal quantitative differences are the higher flexural frequency of the bladder ($f_2 = 13.4$ Hz vs 4.0 Hz) and the smaller coupling ratio (∼7,600 vs ∼66,000). The higher f_2 is a direct consequence of the smaller radius: the stiffness-to-mass ratio scales roughly as $1/R^2$, and the bladder is approximately four times smaller than the abdomen. The smaller coupling ratio reflects the fact that the bladder's f_2 sits above the pelvic transmissibility peak, so the mechanical pathway receives less amplification from the body's own resonance structure.

4. Discussion

4.1. Clinical Implications

The predicted $n = 2$ frequency range (12 Hz to 18 Hz) has immediate relevance to occupational health. Vehicle cabins expose operators to broadband vibration spanning 1 Hz to 20 Hz, with dominant energy typically concentrated below 10 Hz from engine and road excitation [3, 13]. The bladder's fundamental mode sits at the upper edge of this band. Two coupling mechanisms are therefore relevant:

Sub-resonant forced response.. At the ISO 2631 pelvic resonance peak (4 Hz to 8 Hz), the bladder responds as a forced sub-resonant oscillator. The pelvic transmissibility is at its maximum ($T \approx 2.3$ at 5.5 Hz), and while the bladder modal amplification is below unity at these frequencies, the product $T \times H$ remains significant. This is the regime most relevant to occupational WBV exposure: the bladder wall undergoes forced cyclic deformation driven by amplified pelvic motion, even though the excitation frequency is below the bladder's own resonance.

Near-resonant amplification.. At the bladder's natural frequency (12 Hz to 18 Hz), the modal amplification reaches its peak ($Q = 2.5$), but pelvic transmissibility has rolled off ($T \approx 0.3$). This regime may be relevant to certain industrial vibration sources (e.g., pneumatic tools, vibratory compactors) that produce significant energy above 10 Hz. The fill-volume-dependent nature of the resonance means that the bladder's susceptibility to a particular frequency shifts as it fills—a dynamic vulnerability window.

Stretch receptor activation.. Urgency sensation requires activation of stretch receptors (mechanosensitive afferent nerves) in the detrusor muscle and suburothelium [14]. The critical question is whether vibration-induced cyclic wall displacement reaches the activation threshold of these receptors. Our model provides the wall displacement per unit input; coupling this to afferent nerve

thresholds—which are not well-characterised in the dynamic regime—remains an open problem.

4.2. Comparison with Other Pelvic Organs

The bladder is not the only pelvic organ exposed to WBV. The prostate, uterus, and rectum are all fluid-containing or fluid-adjacent structures embedded in the pelvic floor. The present framework could, in principle, be adapted to any of these organs by substituting appropriate geometric and material parameters. The bladder is simply the cleanest example: a single-chamber, fluid-filled, roughly spherical shell with well-characterised mechanical properties.

4.3. Therapeutic Whole-Body Vibration

WBV at 20 Hz to 40 Hz is increasingly used for pelvic floor rehabilitation in post-prostatectomy and postpartum patients [15]. The bioelectrical activity of pelvic floor muscles increases during synchronous WBV [16], and clinical trials report improvements in continence symptoms [17]. Our model suggests that these therapeutic frequencies (20 Hz to 40 Hz) lie above the bladder's $n = 2$ mode for most fill states, placing the bladder in a supra-resonant regime where wall displacement decreases with frequency. The therapeutic effect is therefore more likely mediated by pelvic floor muscle activation than by direct bladder wall resonance.

4.4. Limitations

Several simplifications warrant discussion.

Spherical geometry.. The real bladder is slightly oblate and irregularly shaped, particularly at low fill states. A more realistic geometry would modify the mode shapes and could split the degenerate frequencies of the spherical model. However, at moderate-to-high fill the spherical approximation is reasonable, and the errors are of order $(1 - c/a)^2$ for the eigenfrequencies.

Pelvic constraint.. Approximately 40% of the bladder surface contacts the pelvic bone and pelvic floor musculature. A partial rigid boundary would modify the effective mode shapes and could lower the resonant frequency by restricting the free surface area. This effect is not captured in the free-shell model and represents the most significant limitation of the present analysis.

Nonlinear elasticity.. The exponential strain-stiffening model (equation 6) is a phenomenological fit to vibrometry data. A hyperelastic constitutive model (Mooney–Rivlin or Ogden) would be more physically appropriate for large deformations, though for the small-amplitude vibrations considered here, a linearised modulus is adequate.

Layered wall structure.. The bladder wall comprises distinct layers (mucosa, submucosa, detrusor muscle, serosa) with different mechanical properties. The present model uses a single effective modulus. A multilayer shell model would capture the bending–stretching coupling between layers, potentially modifying both frequencies and damping.

Surrounding tissue.. The bladder is embedded in the pelvic cavity, surrounded by fat, connective tissue, and adjacent organs. This surrounding medium would add both damping and mass loading, likely shifting the resonant frequencies downward and broadening the peaks. The free-shell model therefore represents an upper bound on the resonant frequencies.

5. Conclusions

We have modelled the human urinary bladder as a fluid-filled viscoelastic spherical shell and computed its flexural mode frequencies across the physiological fill range. The principal findings are:

1. The fundamental flexural mode ($n = 2$) lies between 12 Hz and 18 Hz for fill volumes of 50 mL to 500 mL. A frequency minimum of approximately 12 Hz occurs near 170 mL, arising from the competition between geometric

softening (increasing radius and thinning wall) and strain-stiffening of the detrusor muscle.

2. Mechanical vibration transmitted through the pelvic skeleton is approximately 7,600 times more effective than airborne sound at exciting the bladder's $n = 2$ mode at 300 mL fill. This coupling disparity, driven by the $(ka)^2$ penalty in the Rayleigh regime ($ka \approx 0.01$), confirms that whole-body vibration is the dominant excitation pathway—consistent with the epidemiological evidence and with the analogous result for the abdominal cavity.
3. The bladder's predicted resonance range (12 Hz to 18 Hz) sits above the ISO 2631 peak pelvic transmissibility band (4 Hz to 8 Hz). At typical occupational WBV frequencies, the bladder therefore responds as a forced sub-resonant oscillator driven by amplified pelvic motion, rather than being excited at its own natural frequency.
4. The breathing mode ($f_0 \approx 9500$ Hz) is dominated by the fluid bulk modulus and is irrelevant to WBV or infrasonic exposure, confirming the same qualitative separation between breathing and flexural mode families observed in the larger abdominal cavity.

The model provides a first-principles basis for understanding vibration-induced bladder symptoms and may inform both occupational exposure guidelines and the design of therapeutic WBV protocols for pelvic floor rehabilitation. Future work should address the pelvic boundary constraint, validate against finite element analysis with anatomically realistic geometry, and couple the mechanical model to afferent nerve activation thresholds to predict urgency onset directly.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at `https://github.com/jonathan-mace/browntone`.

CRediT authorship contribution statement

Jonathan Mace: Conceptualisation, Methodology, Software, Supervision, Writing – Review & Editing. **Brian R. Mace**: Methodology, Validation, Writing – Review & Editing. **GitHub Copilot**: Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualisation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.